CAPSTONE PROJECT

PROJECT TITLE: INTERACTIVE RECIPE FINDER

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OUTLINE

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- System Development Approach
- Algorithm & Deployment
- Result
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- Future Scope
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PROBLEM STATEMENT:

THE CHALLENGE OF MODERN RECIPE DISCOVERY AND MEAL PLANNING

Home cooks face significant frustration and inefficiency when planning meals due to the inability of current recipe platforms to dynamically account for on-hand ingredients, specific dietary constraints, and limited cooking time.

- **Ingredient Waste:** Cooks often abandon recipes or make time-consuming grocery trips because they cannot easily find meals that utilize their existing pantry inventory.

- **Lack of Personalization:** Existing search filters are often rigid and slow, failing to provide instant customized suggestions that blend multiple criteria (e.g., "Gluten-Free," "Under 30 Minutes," and "Use Chicken Thighs").

- **Decision Fatigue:** The sheer volume of static search results leads to overwhelm and unnecessary time spent sifting through irrelevant options instead of actively cooking.

SYSTEM APPROACH

TECHNOLOGY STACK & DESIGN PRINCIPLES

Core Frontend Technologies

- HTML5: Serves as the foundational structure and semantic organization of the web application.
- CSS3: Utilized for all styling, visual aesthetics, and implementing responsive layouts across all devices.
- JavaScript (ES6+): Enables dynamic interactivity for key user functions, including
- product filtering, shopping cart management, and real-time total calculations.
- **II. Design**
- User Interface (UI): Focused on a clean, intuitive, and user-friendly interface to maximize ease of use.
- Mobile-First Design: Ensures the application is fully mobile-responsive,
- providing an optimal viewing and interaction experience on smartphones and tablets.
- Modular Separation: Adheres to best practices by maintaining distinct, separate files
- for HTML (structure), CSS (style), and JavaScript (logic) for cleaner code and easier maintenance.

Algorithm and Deployment

```
Start
  ↓
Requirement Analysis
  ↓
Project Setup (Git + npm)
  ↓
UI/UX Design (Wireframes & Prototype)
  ↓
Frontend Development (HTML, CSS, JS/React)
  ↓
Frontend Testing
      ├── Unit Testing
      ├── UI/UX Testing
      └── Cross-Browser Testing
  ↓
Optimization (Performance + Responsiveness)
  ↓
Build for Production
  ↓
Deployment (Netlify / Vercel / AWS / etc.)
  ↓
Post-Deployment Testing & Monitoring
  ↓
End
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Magic Recipe Finder</title>
  <link rel="stylesheet" href="style.css"/>
</head>
<body>
  <div class="page-wrapper">
    <nav>
      <h1>🍽️ Magic Recipe Finder</h1>
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="login.html">Login</a></li>
        <li><a href="register.html">Register</a></li>
        <li><a href="about.html">About</a></li>
      </ul>
    </nav>

    <div class="content">
      <section class="slider">
        <div class="slides">
          <div class="slide"><img src="images/Garden-Salad.jpg" alt="Garden Salad"/></div>
          <div class="slide"><img src="images/elements-of-cooking-flavor-profiling.webp" alt="Flavor Profiling"/></div>
          <div class="slide"><img src="images/entree-Hearty-Beef-Vegetable-Soup.jpg" alt="Beef Soup"/></div>
        </div>
      </section>

      <section class="search-box">
        <h2>Enter Ingredients</h2>
        <input type="text" id="ingredients" placeholder="e.g., tomato, cheese, pasta"/>
        <button onclick="findRecipe()">Find Recipe</button>
      </section>

      <section id="recipe-result" class="recipe-display"></section>

<section class="search-box">
  <h2>Enter Ingredients</h2>
  <input type="text" id="ingredients" placeholder="e.g., tomato, cheese, pasta"/>
  <button onclick="findRecipe()">Find Recipe</button>
</section>

<section id="recipe-result" class="recipe-display"></section>

<section class="recipe-slot">
  <h2>🔥 Featured Recipes</h2>
  <div class="recipe-grid">
    <div class="recipe-card">
      <img class="recipe-card img" src="images/Cheesy-Tomato-Pasta-2.jpg" alt="Cheesy Tomato Pasta" />
      <h3>Cheesy Tomato Pasta</h3>
      <p>Rich and creamy pasta with garlic and herbs.</p>
    </div>
    <div class="recipe-card">
      <img class="recipe-card img" src="images/veggie-sandwich-.avif" alt="Veggie Sandwich" />
      <h3>Veggie Sandwich</h3>
      <p>Fresh veggies layered with cheese and mayo.</p>
    </div>
    <div class="recipe-card">
      <img class="recipe-card img" src="images/fruit salad delight.webp" alt="Fruit Salad Delight" />
      <h3>Fruit Salad Delight</h3>
      <p>Colorful mix of fruits drizzled with honey.</p>
    </div>
    <div class="recipe-card">
      <img class="recipe-card img" src="images/Paneer-kathi-Roll-Featured-1.jpg" alt="Spicy Paneer Wrap" />
      <h3>Spicy Paneer Wrap</h3>
      <p>Grilled paneer and veggies in a tortilla.</p>
    </div>
  </div>
  <button onclick="addRecipe()" class="add-recipe-btn">➕ Add New Recipe</button>
</section>
```

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RESULT

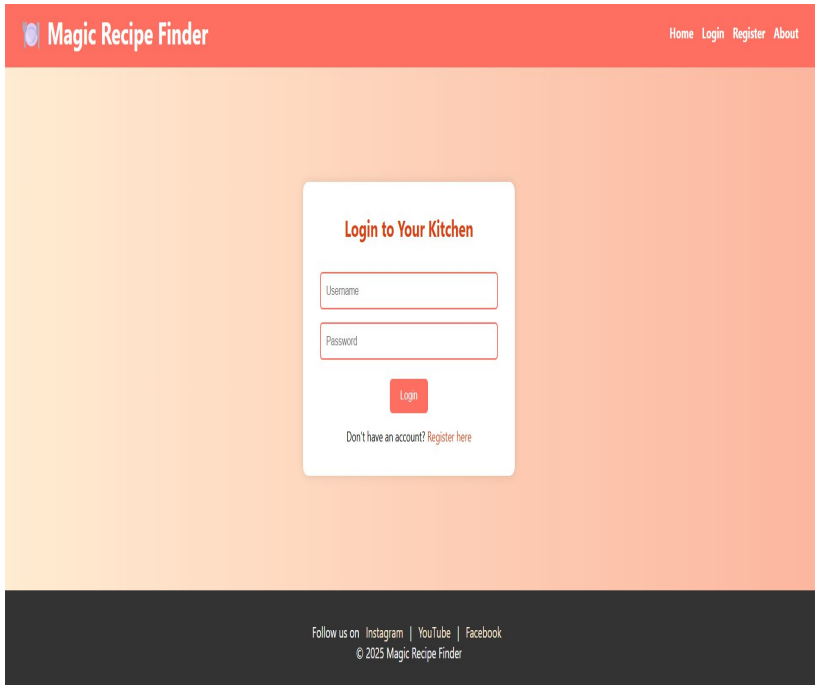
```html
<html lang="en">
<head>
  <style>
    nav {
      background: #ff6f61;
      color: white;
      padding: 1em 2em;
      display: flex;
      justify-content: space-between;
      align-items: center;
    }

    nav ul {
      list-style: none;
      display: flex;
      gap: 1em;
    }

    nav a {
      color: white;
      text-decoration: none;
      font-weight: bold;
      transition: color 0.3s ease;
    }

    nav a:hover {
      color: #ffe0b2;
    }

    .register-box {
      background: white;
      padding: 2em;
      border-radius: 10px;
      box-shadow: 0 0 10px rgba(0,0,0,0.1);
      text-align: center;
      max-width: 500px;
      width: 100%;
```



Git hub link: https://hemanthbinnala.github.io/Recipe-finder/

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CONCLUSION

- The Problem Solved
- The complexity and cost of traditional e-commerce solutions currently prevent small
- businesses from establishing effective online product displays.
- The Solution's Impact
- Empowerment: We deliver a lightweight, visually appealing, and
- cost-effective platform that instantly empowers small businesses to showcase their products digitally.
- Customer Experience: The use of JavaScript-driven interactivity
- filtering, real-time cart) ensures a modern, user-friendly, and engaging customer browsing experience.
- Final Takeaway
- By focusing on simplicity, modularity, and essential frontend technologies, our solution bridges
- the digital gap, allowing small businesses to compete effectively online without the burden of complexity or high

FUTURE SCOPE(OPTIONAL)

- Persistent Data Storage: Implement local storage (Browser-based) or a simple,
- free Backend-as-a-Service (BaaS) like Firebase or Supabase to save cart data and product updates permanently.
- Simple Authentication: Introduce basic user sign-in/sign-up, enabling features
- like saving Wishlist's and tracking order history (if an order process is added).
- Advanced Filtering: Extend product filtering beyond category to include Price Range sliders
- and Multi-select attribute filters (e.g., color, size, material).

REFERENCES

Reference-1

Mozilla Developer Network (MDN) Web Docs

- NanthiniMoha/TNSDC-FWD-DIGITAL_PORTFOLIO
- Get started with Bootstrap · Bootstrap v5.3

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THANK YOU

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